Friedrich-Alexander-Universität
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Naturwissenschaftliche Fakultät

Lehrstuhl
Mathematik und
Data Science

PROJECT PRESENTATION

Hybrid AI Framework for Employee Project Allocation (HAF-EPA)

An intelligent and hybrid approach combining machine learning and optimization techniques to achieve fair, efficient and skill-aligned project allocation.

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Project Repository
https://github.com/firozfau/HAF-EPA

Dataset
https://www.kaggle.com/datasets/firozfau/software-development-employee-project-dataset

HAF-EPA ?

Hybrid AI Framework for Employee Project Allocation (HAF-EPA)

An intelligent decision-support system designed to recommend the most suitable employees for project assignments by analysing workforce data from multiple perspectives.

1. Core Components of HAF-EPA

Machine Learning:
Predicts employee suitability based on historical workforce data.

Knowledge Graph:
Represents relationships among employees, skills, projects, and experience.

Rule-Based Filtering:
Applies practical constraints such as availability, required skills, and project conditions.

Overall Goal:
To generate accurate, transparent, and data-driven employee recommendations for project staffing decisions.

2. Existing Challenges in Employee Project Allocation

As organisations become increasingly project-oriented and workforce datasets continue to grow, assigning the right employees to the right projects has become a complex decision-making task. Traditional employee allocation approaches often face several operational and managerial challenges that can negatively affect project success and workforce utilisation.

Key Challenges:

1. Manual and time-consuming allocation process
2. Subjective decision-making based on human judgement
3. Over-reliance on technical skill matching
4. Poor utilisation of workforce resources
5. Difficulty scaling across large organisations
6. Risk of overlooking highly qualified employees
7. Limited consideration of performance, availability, and collaboration factors
8. Lack of intelligent and data-driven decision support

3. **Impact of Existing Allocation Challenges**

The limitations of traditional employee allocation approaches can significantly affect project outcomes and organisational performance. When staffing decisions rely primarily on manual judgement and simple skill matching, several operational challenges may arise.

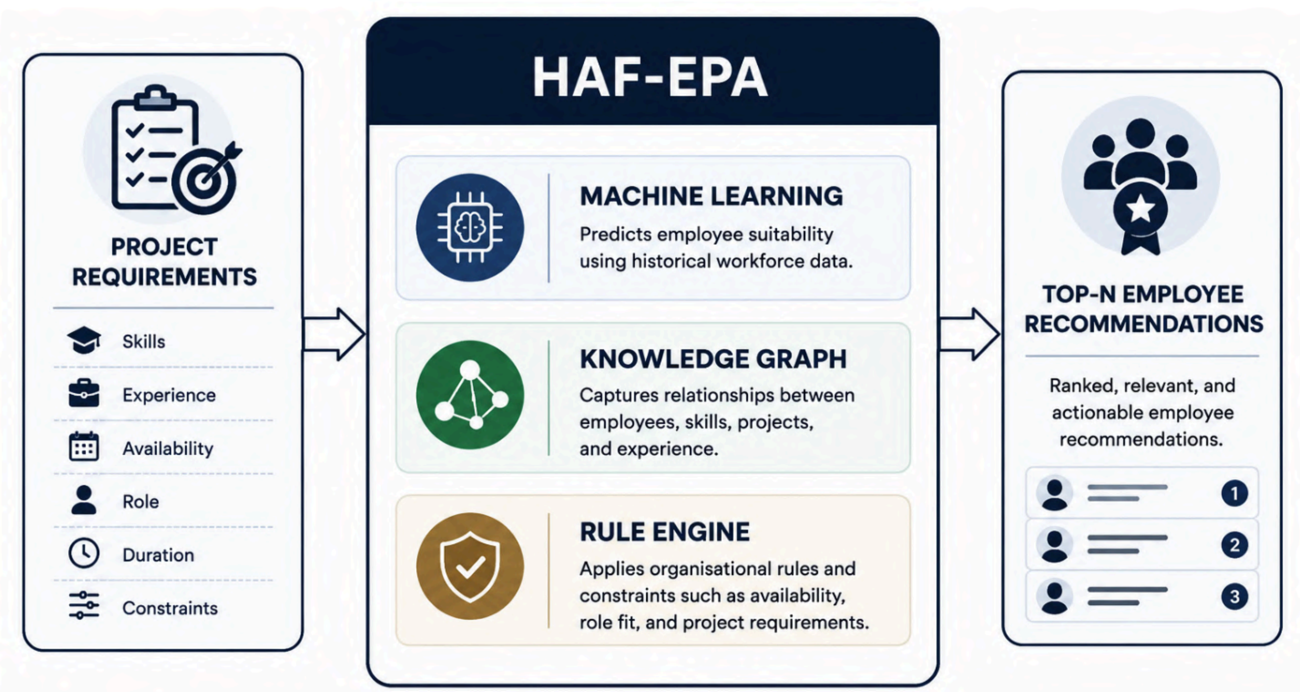
Business Impact:

1. Reduced project success and delivery performance
2. Inefficient utilisation of employee skills and expertise
3. Inconsistent and biased allocation decisions
4. Missed opportunities to identify the best candidates
5. Increased staffing effort and decision-making time
6. Lower organisational productivity and resource efficiency

4. Proposed Solution: HAF-EPA

HAF-EPA is an intelligent employee recommendation framework designed to support project staffing decisions through multi-dimensional workforce analysis.

The framework evaluates employee suitability using workforce capabilities, project requirements, historical performance, availability, and organisational context to generate accurate and data-driven employee recommendations.



Source: Author's own illustration.

5. Research Methodology and System Workflow

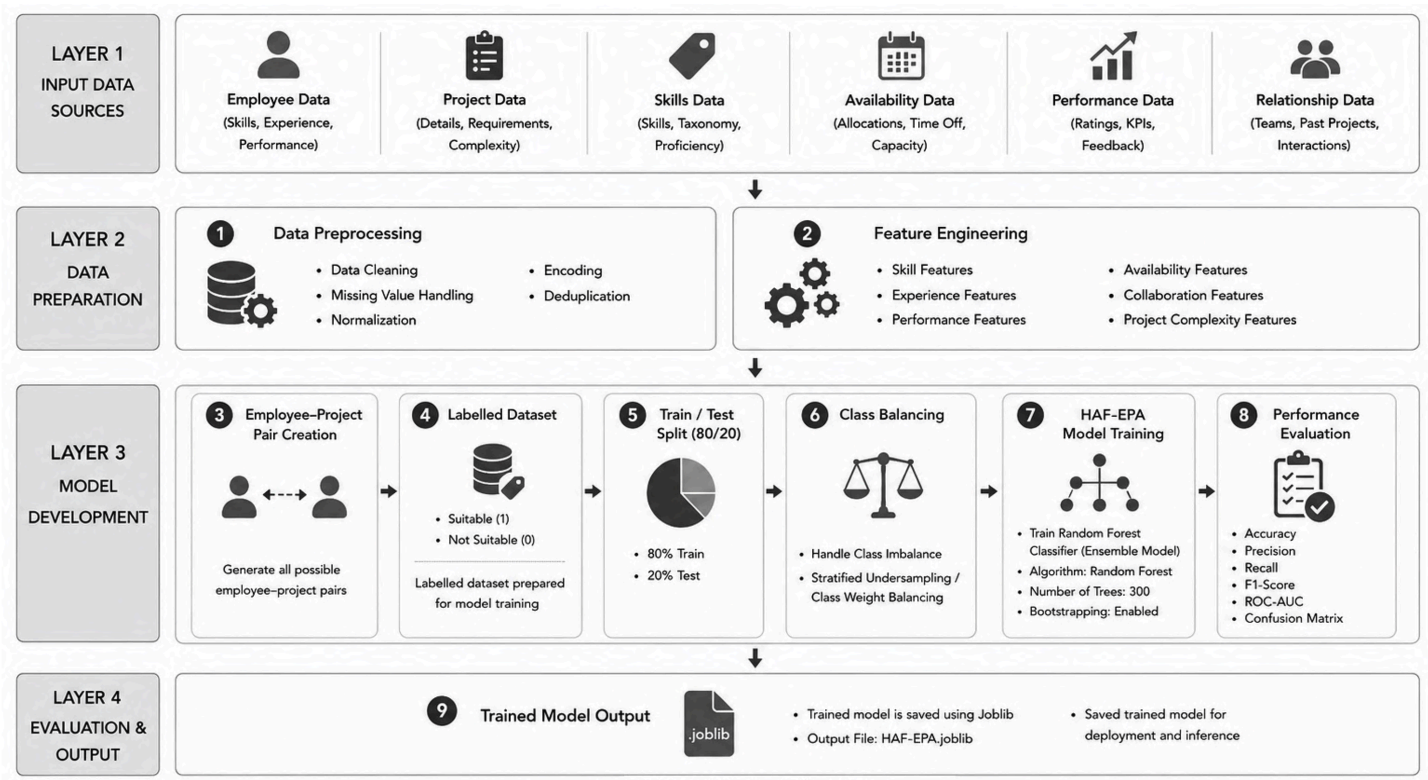


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Figure 3.1: Training Pipeline and Model Development Architecture of HAF-EPA

Multiple workforce datasets are collected and integrated to create a unified workforce repository. Data preprocessing and feature engineering are then performed to generate 19 workforce features, including skill matching, experience alignment, availability fitness, past performance, and collaboration compatibility. These features are used to create employee–project pairs, which are subsequently labelled for supervised learning. The Random Forest model is then trained and validated using an 80:20 train–test split strategy. Finally, the trained model is deployed to generate intelligent employee recommendations and support project staffing decisions.

The overall workflow consists of data collection, feature engineering, employee project pair generation, dataset labelling, model training, validation, and recommendation generation.

6. **Overall Methodology Flow**

Input Data Sources → Data Preparation → Feature Engineering → Pair Generation → Labelling → Train-Test Split → Class Balancing → Random Forest → Validation → Knowledge Graph → Hybrid Recommendation → Top-N Ranking

7. Input Data Sources:

HAF-EPA collects workforce information from multiple sources, including employee data, project data, skills data, availability data, performance data, and relationship data. These datasets provide the foundation for employee suitability analysis.

The integrated datasets include Employee Data, Project Data, Skill Data, Availability Data, Performance Data, and Relationship Data.

8. **Data Preparation:**

Raw datasets are cleaned and transformed into machine-readable format.

In this step, inconsistent and incomplete data are cleaned. Categorical values are encoded and numerical values are standardised so that the data can be used by the machine learning model.

Such as , Data cleaning , Missing value handling , Deduplication , Encoding , Normalisation and Dataset integration

9. Feature Engineering:

Instead of using raw data directly, I created 19 meaningful features such as skill match, weighted skill match, experience score, availability fitness, past performance, and soft-skill compatibility.

Skill Match Score:

SkillMatch(e,p) = |Se ∩ Sr| / |Sr|

Availability Fitness:

AvailabilityFit(e) = 1 - AllocationPercent(e) / 100

Category	Example Features
Skills	Skill Match, Weighted Skill Match
Experience	Experience Score, Project Complexity Fit
Availability	Availability Fitness
Performance	Past Performance Score
Collaboration	Collaboration Score

10. Employee–Project Pair Generation:
Every possible employee–project combination was generated. Each pair represents a potential assignment. Each employee is paired with each project.

Employee × Project = Employee–Project Pair
2500 × 900 = 2.25 million pairs

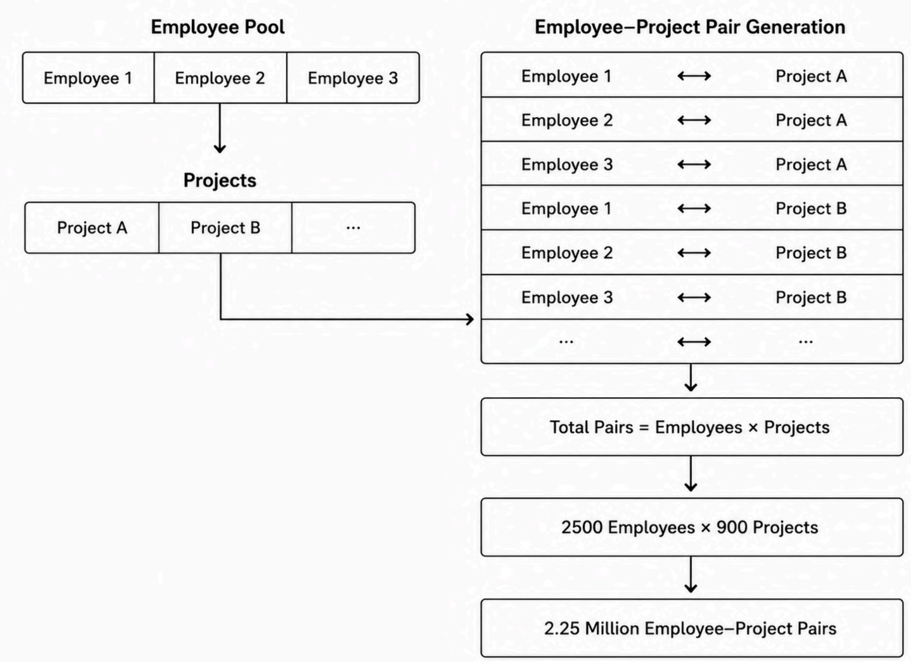
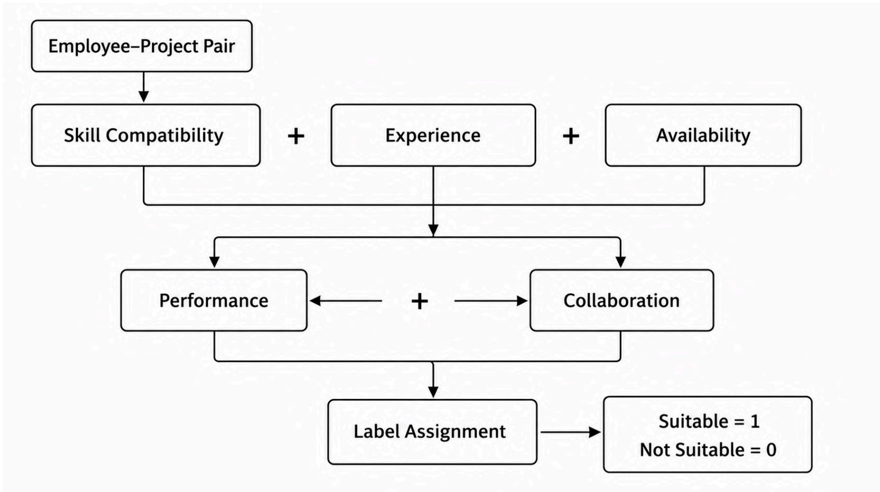


Figure 4.1: Employee–Project Pair Generation Process

11. Dataset Labelling:
Labels were created using skill compatibility, experience, availability, performance, and collaboration factors. Each pair receives a binary label.
Suitability Score = Skill + Experience + Availability +Performance + Collaboration

If Score ≥ Threshold ⟶ Suitable (1)
Else ⟶ Not Suitable (0)



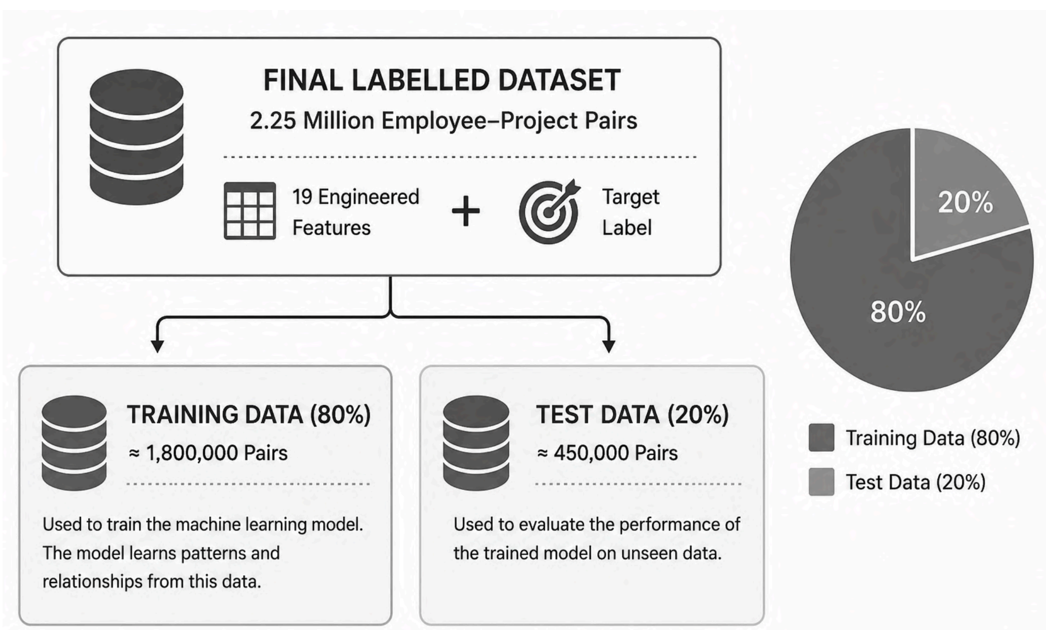
12. Train-Test Split:

The labelled dataset was divided into training and testing datasets.

Dataset → Training Data = 80%   Testing Data = 20%

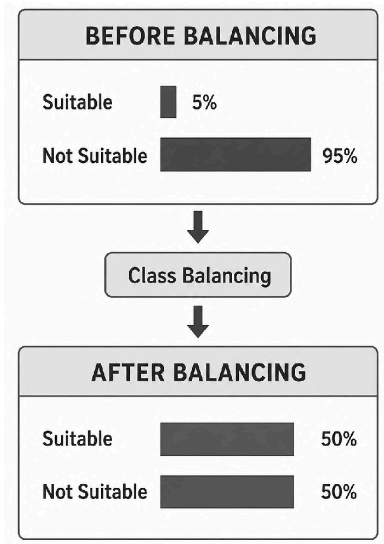
Training: about 1.8 million pairs

Testing: about 450,000 pairs



13. Class Balancing:

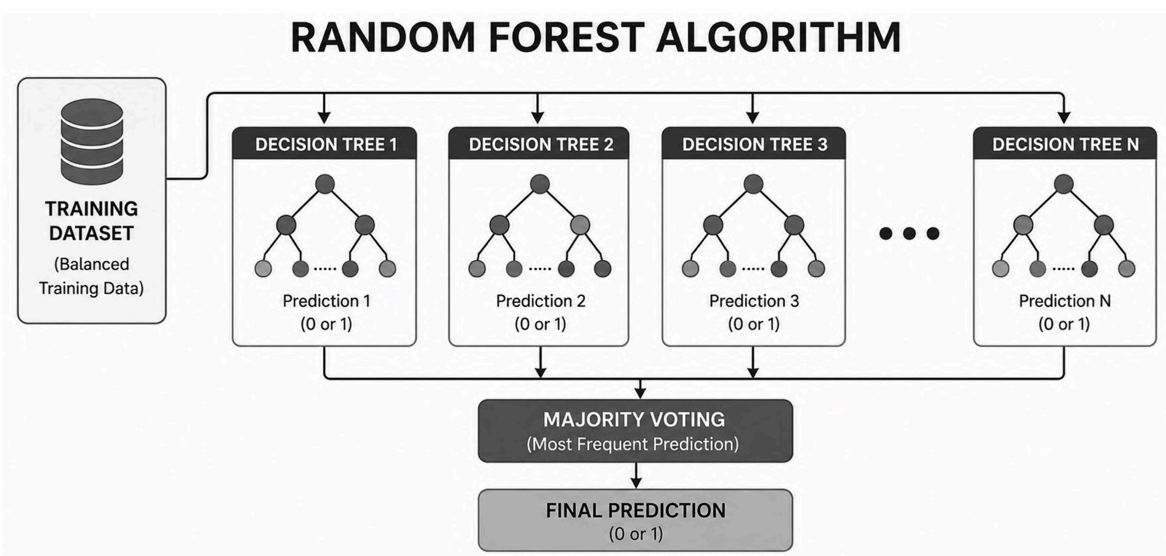
Most employee–project pairs are not suitable. Since suitable employee–project pairs were very few compared to non-suitable pairs, class balancing was applied to reduce model bias. We make solution using, Random undersampling , Class weight balancing.



14. Random Forest Training:

Handles structured data well Reduces overfitting Works with many features Provides feature importance Good for classification.

n_estimators = 300, max_depth = 14, class_weight = balanced, random_state = 42



Why Random Forest?

1. Handles structured workforce data
2. Reduces overfitting
3. Supports feature importance analysis
4. High prediction accuracy
5. Robust for large datasets

15. Model Validation:

trained model + test data → evaluation metrics.

Precision = TP / (TP + FP)   Recall = TP / (TP + FN)

Metric	Value
Precision	2.19%
Recall	93.89%
Accuracy	88.98%
Weighted F1	93.92%

Confusion Matrix:

	Predicted Yes	Predicted No
Actual Yes	TP	FN
Actual No	FP	TN

16. Knowledge Graph:

This helps the system identify employees even when exact skill matching is not enough. Knowledge Graph captures semantic relationships, employees ↔ skills ↔ projects.

Example:

Machine Learning is related to Data Science.
Python is related to Backend Development.
React is related to Frontend Development.

17. Hybrid Recommendation:

ML Score + KG Score + Rule Score → Hybrid Score.

Formula: HybridScore = w1(MLScore) + w2(KGScore) + w3(RuleScore)

Explain:

ML Score = model prediction

KG Score = semantic skill relevance

Rule Score = availability and constraints

18. Top-N Employee Ranking:

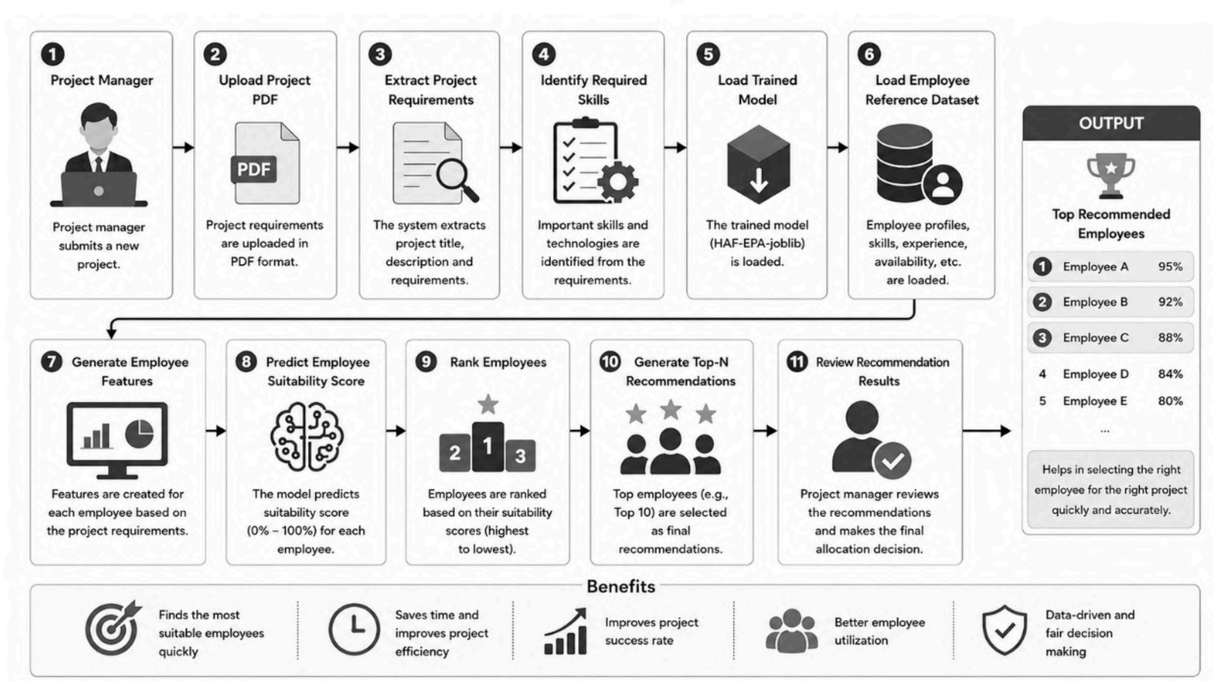
Employees are sorted according to their final hybrid scores. All candidates are ranked and the Top-10 employees are selected.

Higher score = higher recommendation rank

Rank	Employee	Score
1	Emp A	95%
2	Emp B	92%
3	Emp C	89%

19. Web Application Implementation:

The HAF-EPA framework was implemented as a web-based decision-support application using Python, Flask, Scikit-learn, and Chart.js. The application enables project managers to upload project requirements, generate employee recommendations, and visualise recommendation results through an interactive interface.



Source: Author's own illustration.

Figure 5.1: HAF-EPA Web Application Workflow (Real Usage Scenario)

The HAF-EPA web application provides a practical implementation of the proposed recommendation framework. The workflow begins when a project manager uploads a project requirement document in PDF format. The system automatically extracts project information and identifies the required technical and management-related skills. Subsequently, the trained HAF-EPA model and employee reference dataset are loaded.

Based on the extracted project requirements, employee–project features are generated for each candidate employee. These features are evaluated by the trained Random Forest model to predict employee suitability scores. The recommendation engine then ranks employees according to their suitability scores and generates the Top-N recommended candidates. Finally, the project manager reviews the ranked recommendations and selects the most suitable employees for project allocation.

The web application transforms complex workforce data into an interactive and user-friendly decision-support system, enabling faster, more transparent, and data-driven employee allocation decisions.

20. Conclusion:

HAF-EPA is a Hybrid AI-based employee recommendation framework that integrates Machine Learning, Knowledge Graph reasoning, and Rule-Based Filtering to generate intelligent, transparent, and data-driven employee recommendations for project staffing decisions.